



Autumn 2025

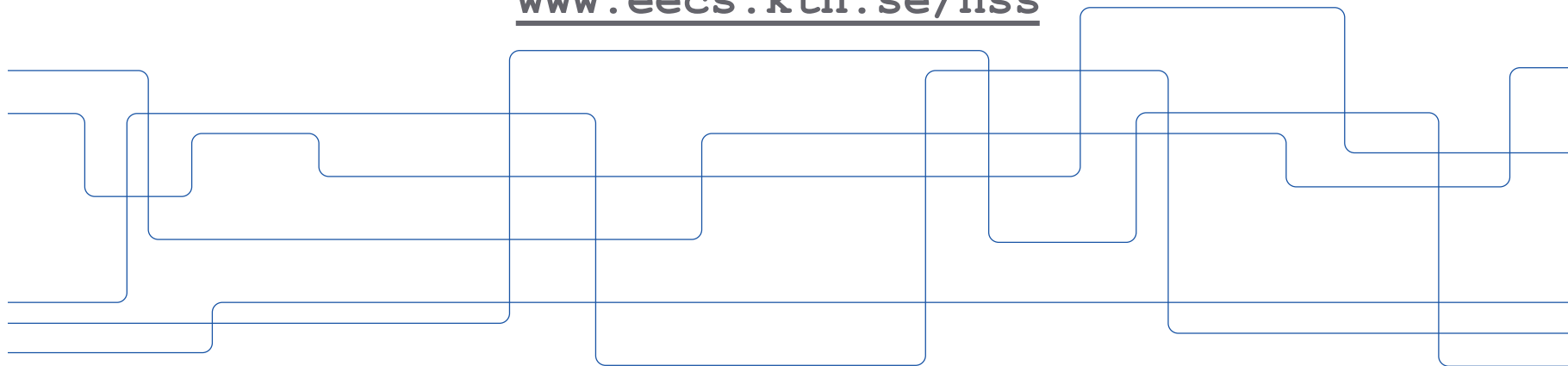
Discussion leads

**Marco Casagrande
& Ahmed Hussain**

Privacy Enhancing Technologies (PETs) for Mobile Networks

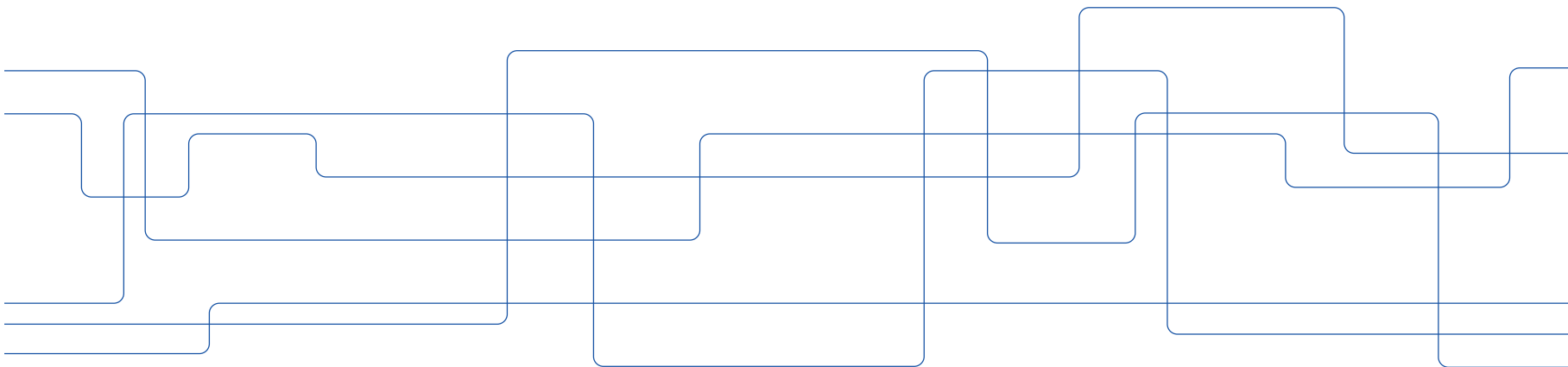
Panos Papadimitratos

www.eecs.kth.se/nss





Introduction



Mobile Networks



Src: <https://www.ssm-gmbh.com/smart-buildings>

Mobile Network Applications

Vehicular Communication Systems



Illustration: C2C-CC

Urban Sensing Systems

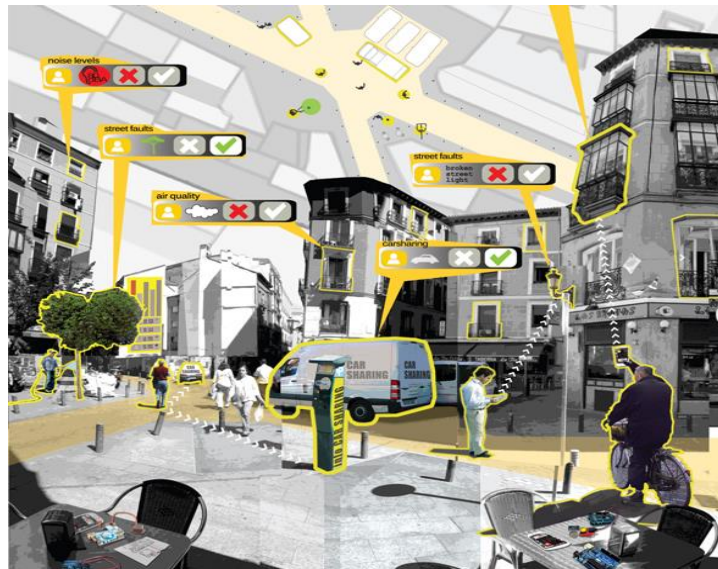


Illustration: complexitys.com

Privacy

ISO/IEC 15408-2

- ISO/IEC 15408-2 identifies 4 key attributes
 - Anonymity – unable to determine the real identities
 - Pseudonymity - accountable use a resource or service without disclosing real identities
 - Unlinkability - multiple uses of resources or services without being linked
 - Unobservability - use a resource or service without others, especially third parties, being able to observe

Anonymity



Ex: Whisper,
Yik Yak

Pseudonymity



Ex: Twitter,
Reddit

Real Identity



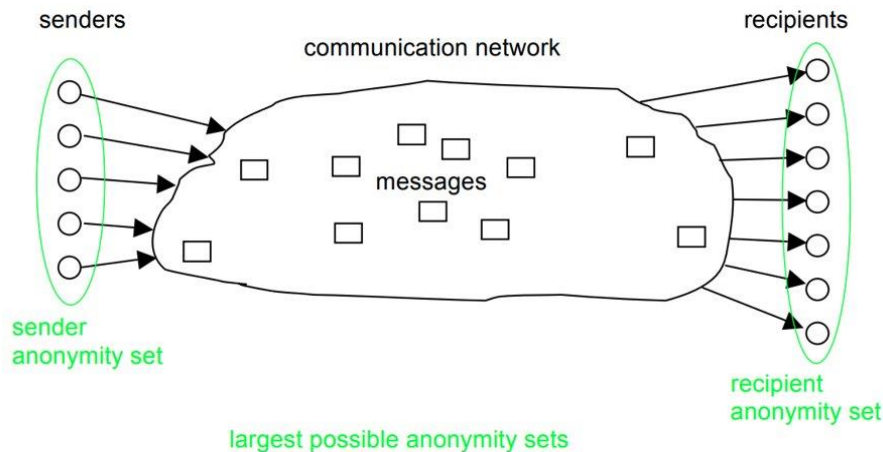
Ex: Facebook,
LinkedIn

[Credit: <https://guides.co/g/the-network-effects-bible/121727>]

Privacy (cont'd)

Anonymity

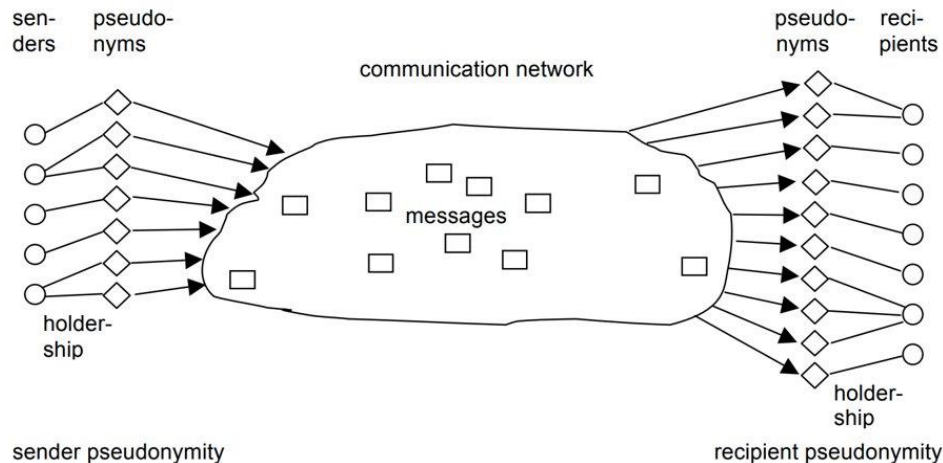
- Remove real identities from messages
- Usually quantified with anonymity set
- Anonymity of a subject means that the subject is not identifiable within a set of subjects, the anonymity set



Privacy (cont'd)

Pseudonymity

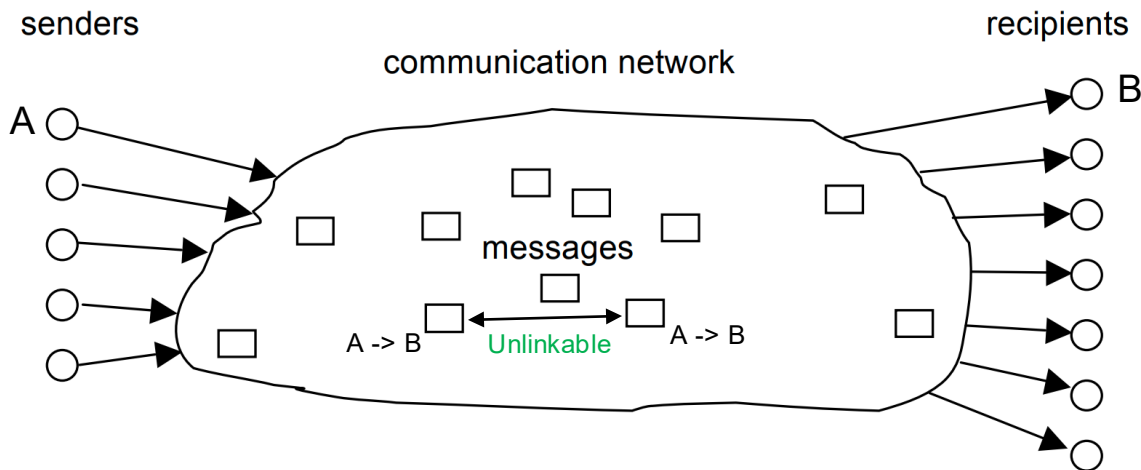
- Replaces with pseudonymous identities
- Conditional and accountable anonymity
 - Reveals user identity upon misusing the anonymity



Privacy (cont'd)

Unlinkability

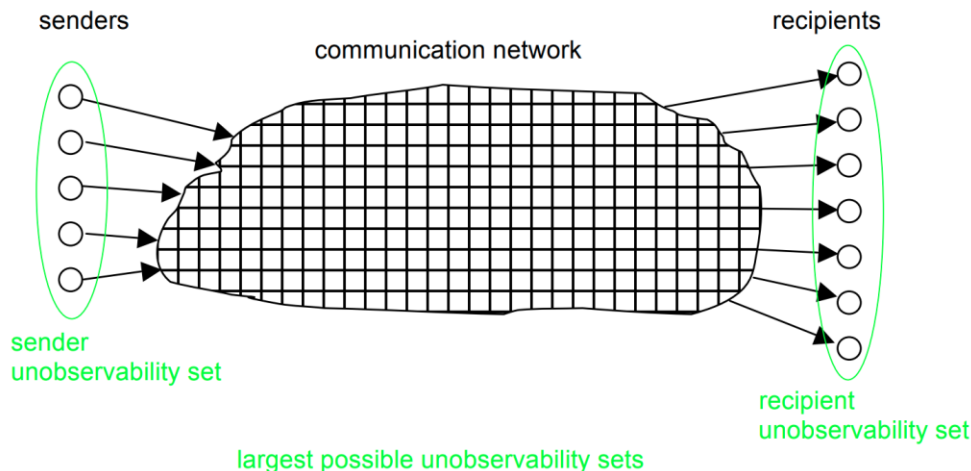
- The attacker cannot sufficiently distinguish whether the user actions are related or not



Privacy (cont'd)

Undetectability and unobservability

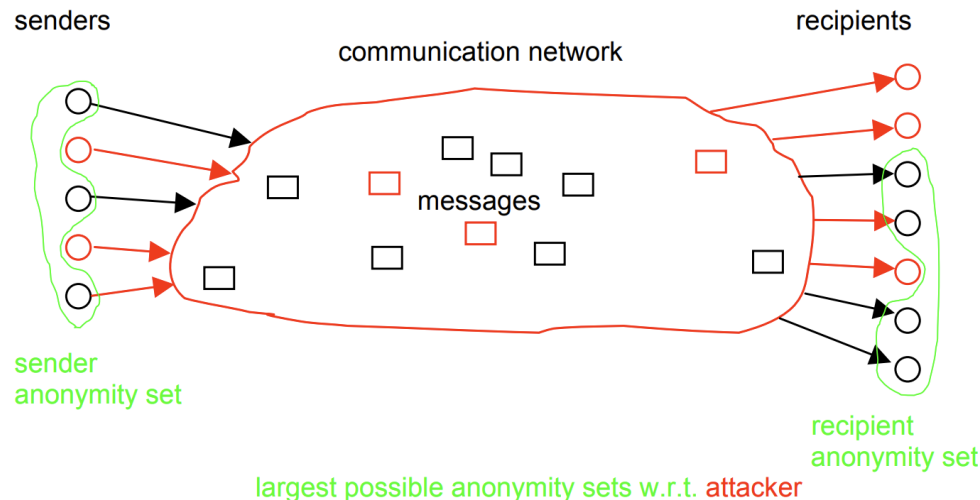
- Undetectability means the attacker cannot sufficiently distinguish whether an action performed or not
- Unobservability means undetectability against all uninvolved subjects



Adversaries

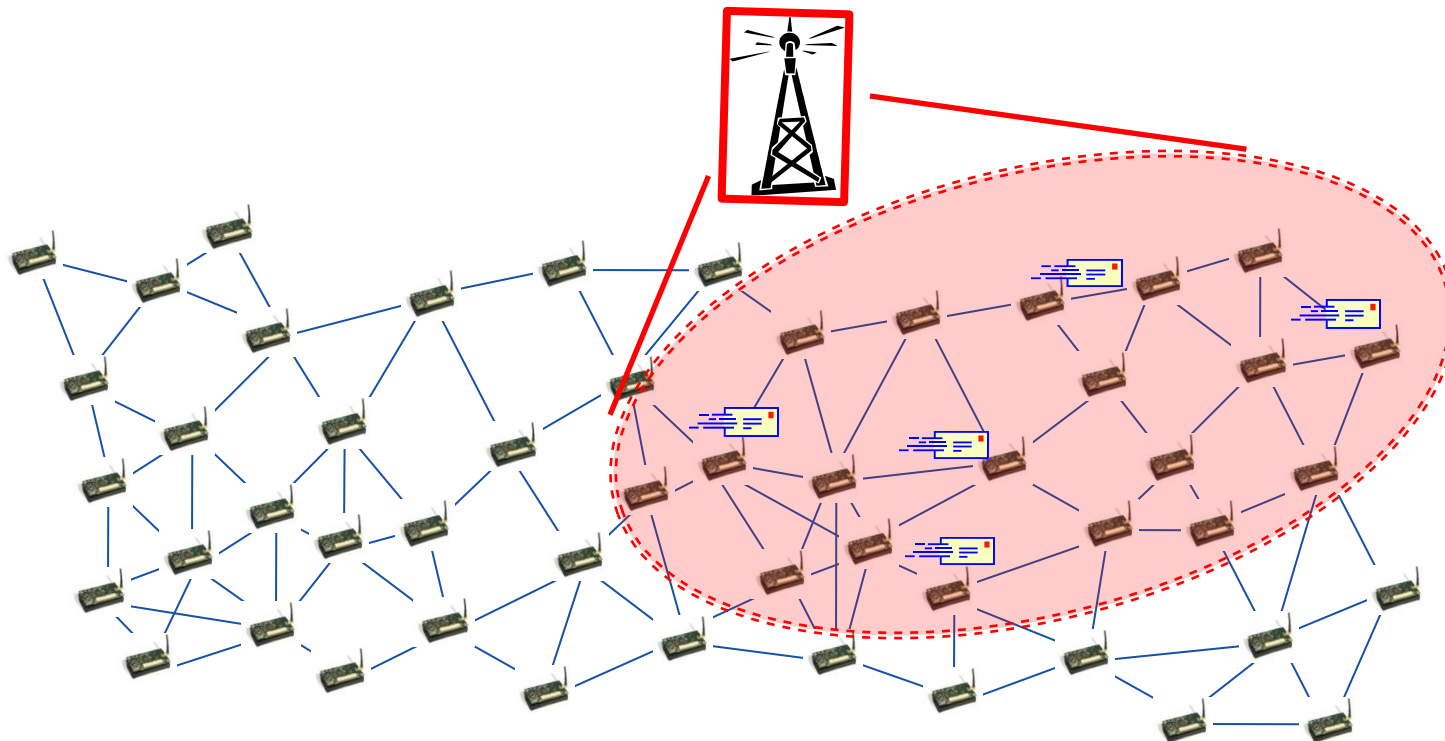
Objectives

- Attacker is a set of entities working against privacy protection goal, including any type of adversaries
- Minimize anonymity sets based on her ability and knowledge



Adversaries (cont'd)

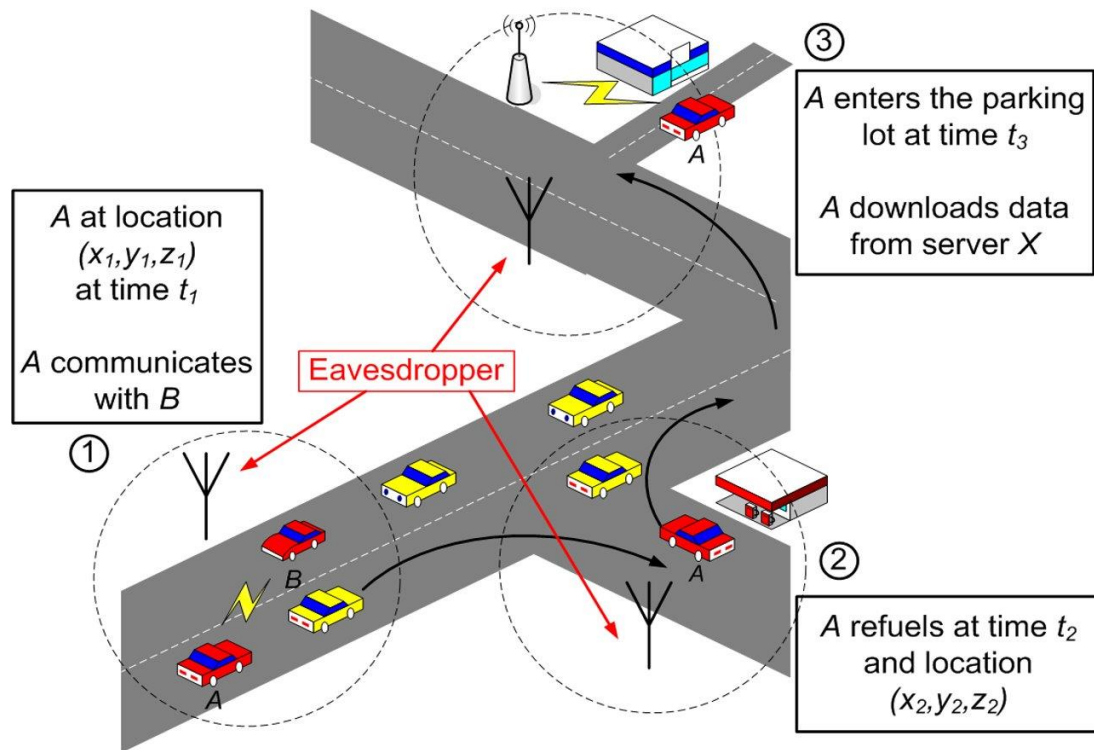
External eavesdroppers



Adversaries (cont'd)

External eavesdroppers

- Regional or global eavesdroppers
- Linking of actions based on identities
- Linking of content semantically
 - Linking of locations based on mobility constraint





Adversaries (cont'd)

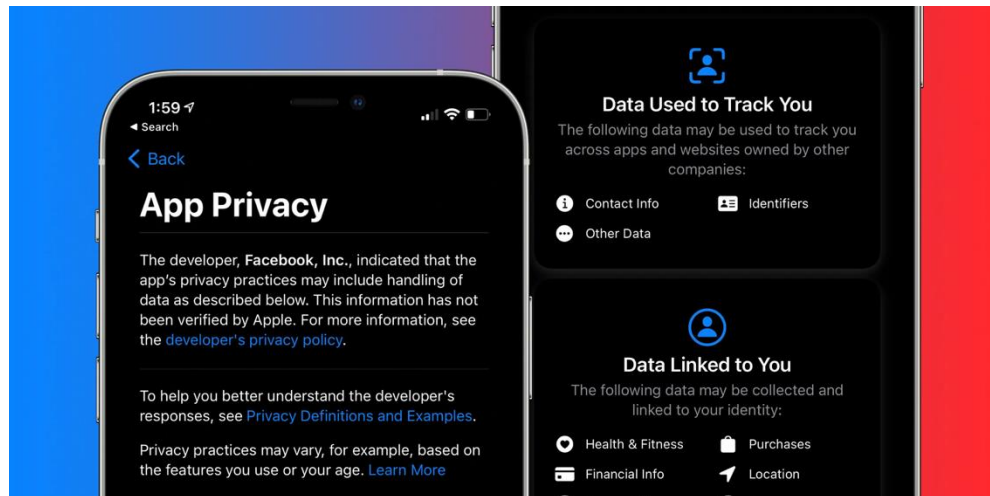
Honest-but-curious entities

- Internal attacks can target user privacy
- Honest-but-curious system entities: they execute correctly the various protocols but are curious to infer user data
- They can collude with other system entities
- Motive: monetization of user privacy

Adversaries (cont'd)

Honest-but-curious service providers

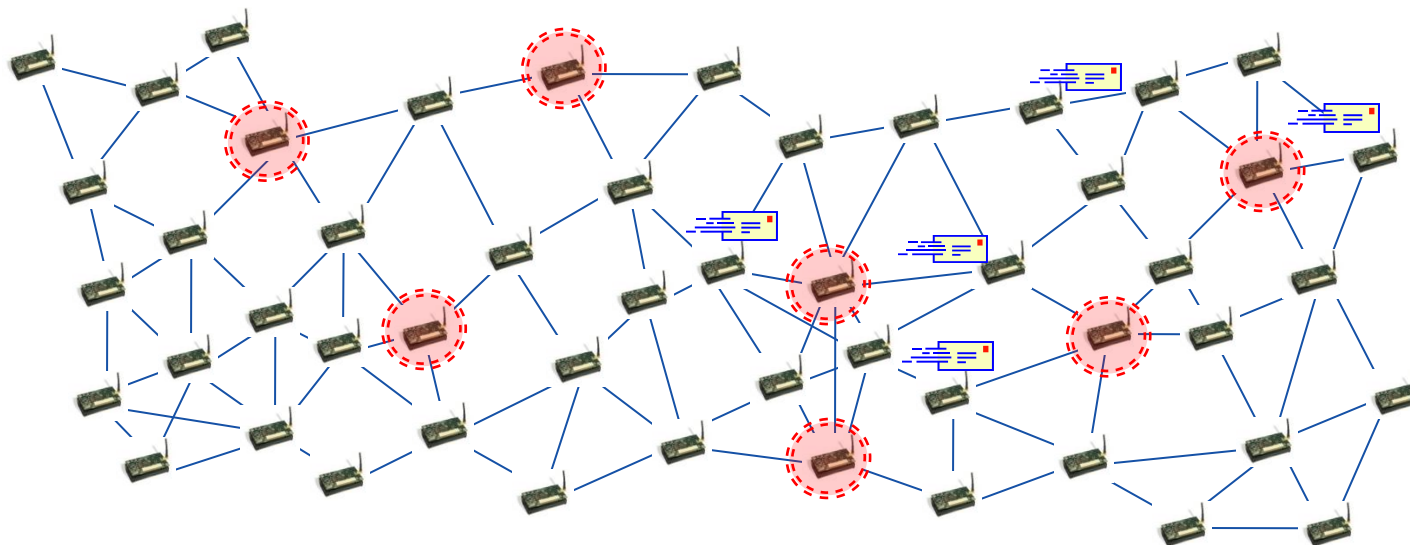
- User actions in the apps are being collected
- Hard to verify provider promises in Privacy Policy



Credit: <https://9to5mac.com/2020/12/17/how-to-check-ios-app-privacy-details-iphone-ipad/>

Adversaries (cont'd)

Honest-but-curious peers

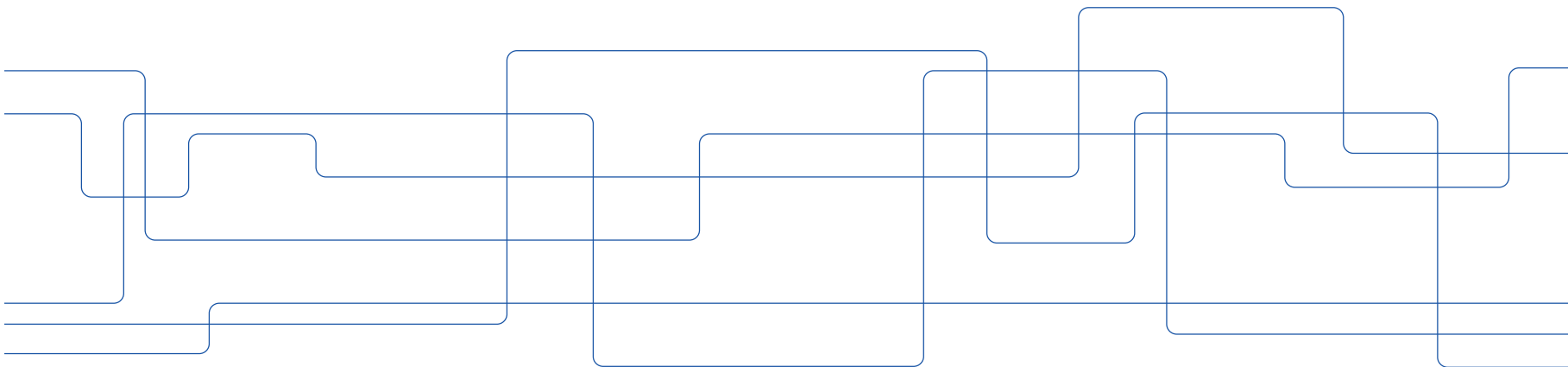




Privacy on top of security

- Security was priority
- Security should not be degraded with privacy protection
- Ano- / Pseudo- nymity with
 - Authentication
 - Accountability
- Are security and privacy at odds?
 - Protect both systems and users from each other

Privacy-preserving Security Credentials



Pseudonymous authentication

- Pseudonym
 - No identifying information in certificate
- Pseudonym format

PNYM

Short-term Cert()

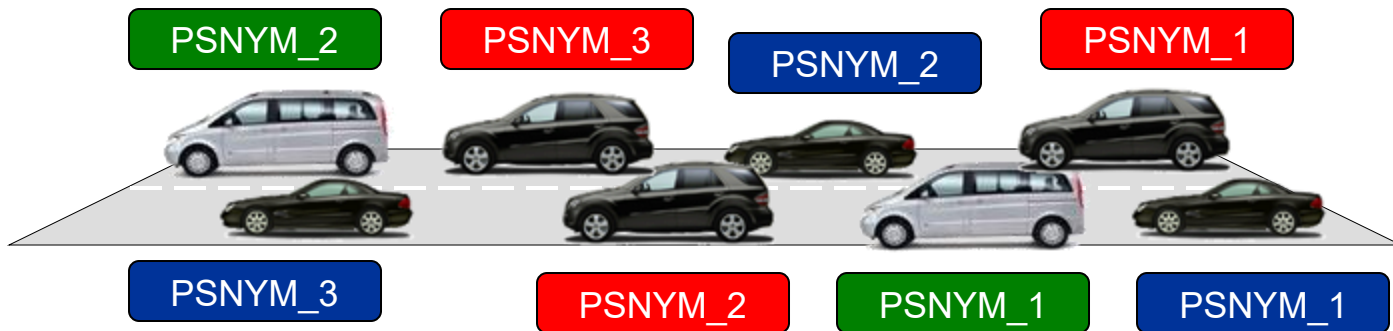
PSNYM-Provider ID	PSNYM Lifetime
Public Key K_i	
PSNYM-Provider Signature	

- Pseudonym provider: a trusted third party

Pseudonymous authentication

Use case - secure Vehicular Communication Systems (VCS)

- Equip vehicles with multiple pseudonyms
 - Alternate among pseudonyms over time (and space)
 - Sign message with the private key corresponding to pseudonym
 - Append current pseudonym to signed message



Group signatures


Group A
public key 
↓
Valid signature from a
legitimate member of
Group A
??? Member ???

Group A



Group member
signing keys



gsk_1



gsk_2



gsk_3



Group manager
generates
signing keys and
a public key

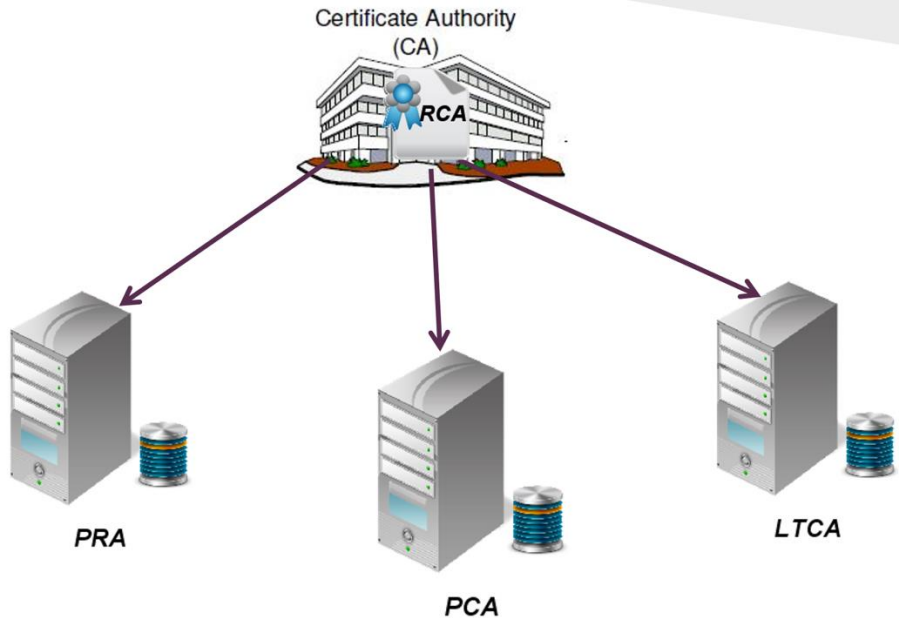


Group signatures (cont'd)

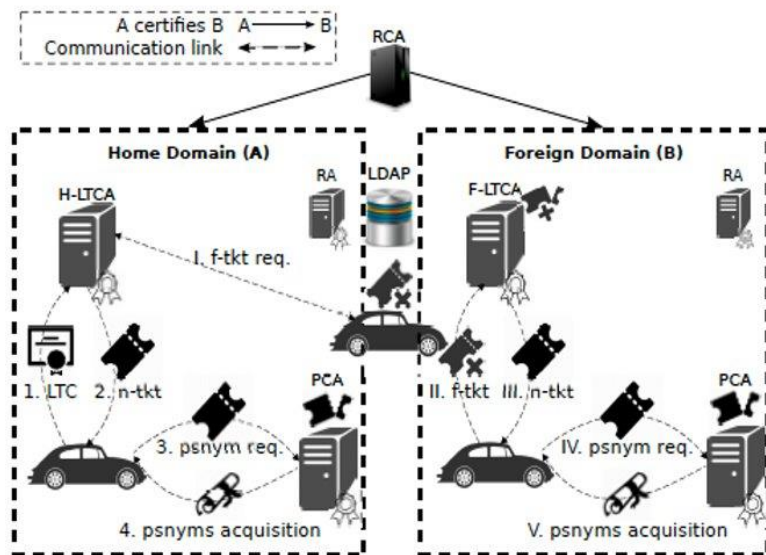
- Authentication
 - Any node can sign on behalf of the group
- Unlinkability
 - Given two messages and their signatures, we cannot tell if the signatures were from the same signer or not
- Traceability and inescapability
 - The group manager can reveal the signer with its secret key
- Scalability
 - Possible but complex to add or remove members once the group is formed
 - Usually required to involve all members for key update/revocation

Privacy against Credential Providers

- Users need to authenticate to the infrastructure to obtain the privacy-preserving credentials
 - Linkability from user identifiable information to the credentials
- Duty separation
 - Credential issuance involves multiple infrastructure entities
 - Limit the knowledge each entity learns

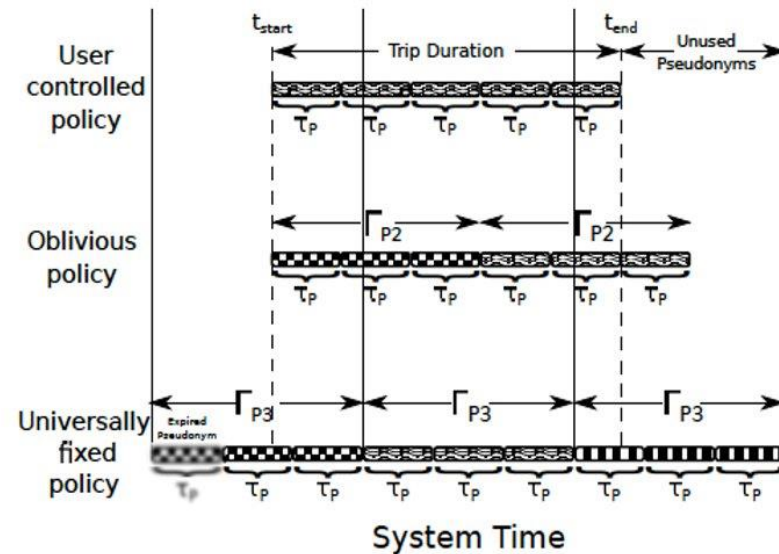


SECMACE



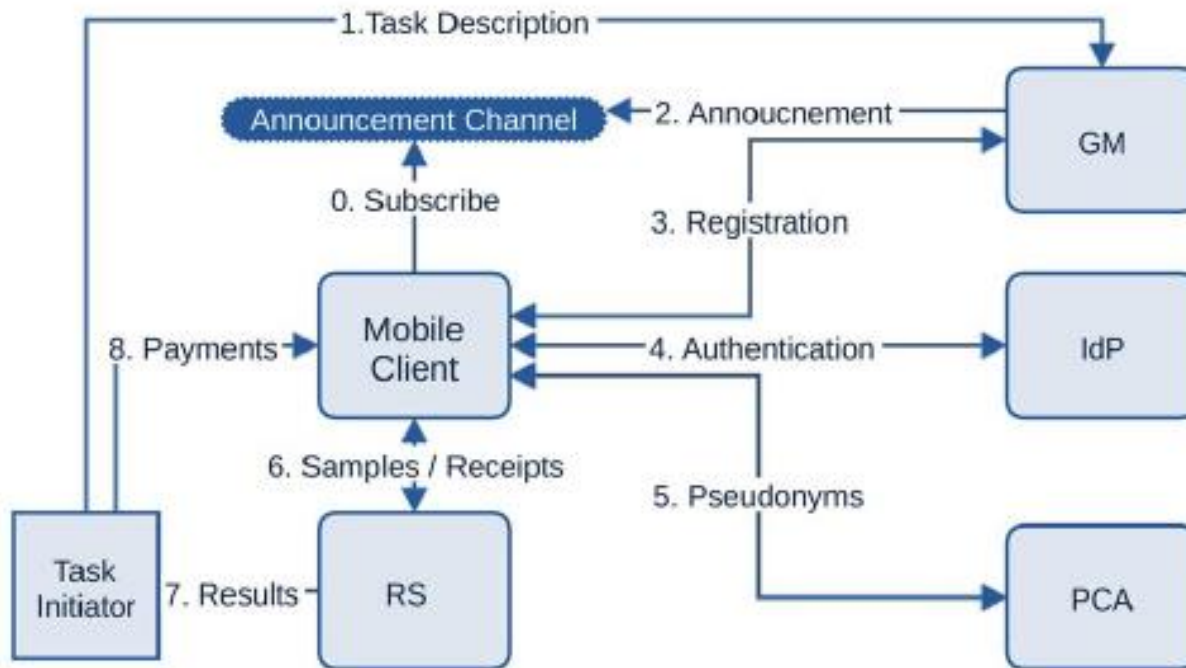
Pseudonym acquisition overview in the home and foreign domains.

[KhodaeiJP:J:2018]



Pseudonym Acquisition Policies.

SPPEAR

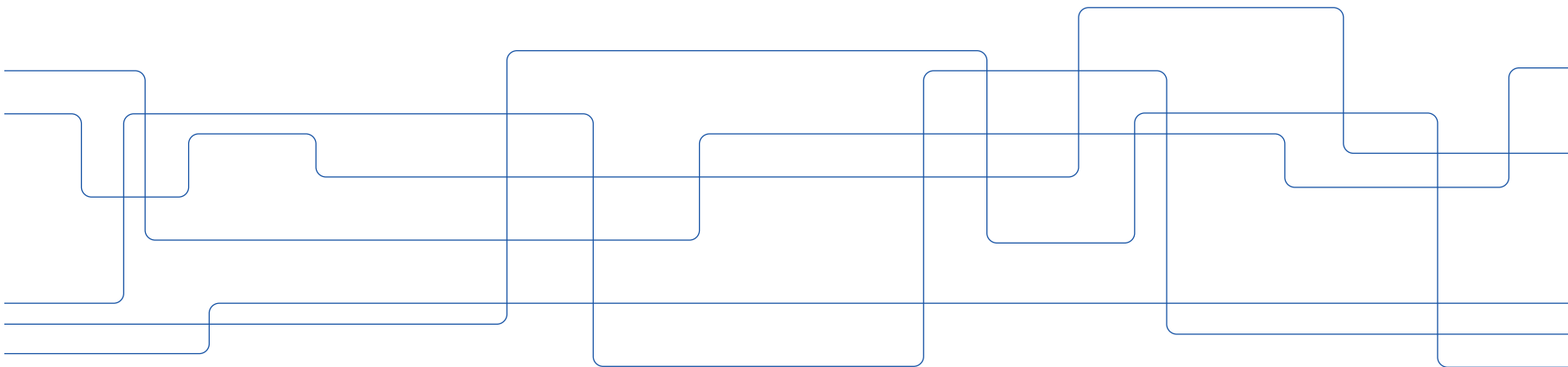


Honest-but-curious SPPEAR Entities

Honest-but-curious (colluding) entities	Information Leaked	Privacy Implications
GM	-	No sensitive information can be inferred.
IdP	t	The IdP can simply infer that an anonymous user wishes to participate in a task.
PCA	PS, t	The PCA will infer that an anonymous user wishes to receive pseudonyms for a given task.
SAS	s, PS, r	The SAS knows that a given report was submitted for a specific sensing task.
GM, IdP	t, id	The GM and the IdP can infer that a user with a known identity wishes to participate in a specific task.
GM, PCA	t, id, PS	The GM and the PCA can infer that a user with a known identity wishes to participate in a specific task and has received pseudonyms.
GM, SAS	s, PS, r	When the GM and the SAS collude they can infer that a report was submitted by a pseudonymous user.
IdP, PCA	t, PS	These authorities can infer that an anonymous user received pseudonyms for a specific task.
PCA, SAS	t, PS, s, r	The PCA and the SAS can infer that an anonymous user received pseudonyms for a specific task and has submitted a report.
GM, PCA, SAS	all	Full de-anonymization of the user, the task she participates in and the reports she has submitted.

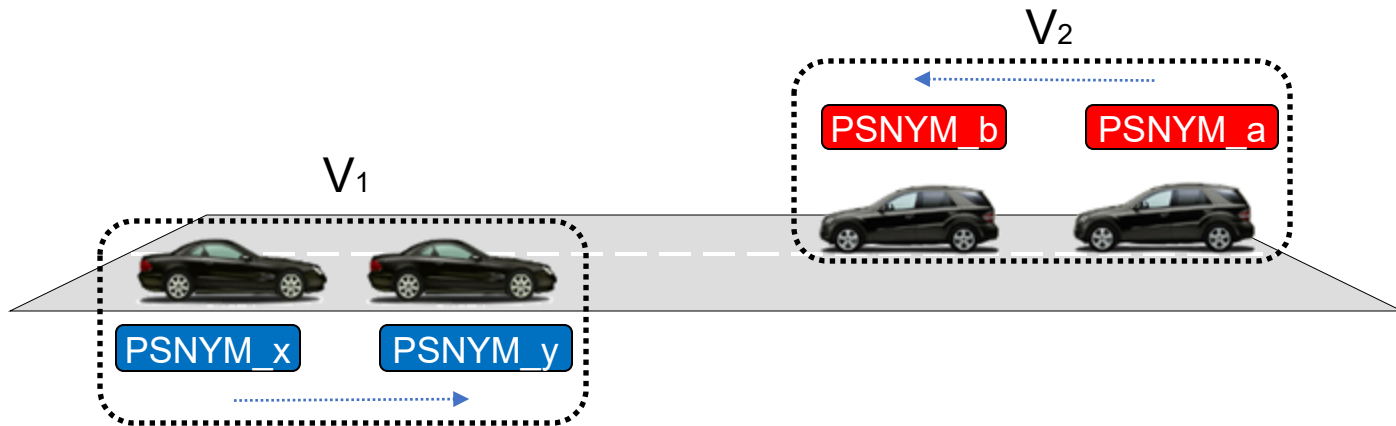
Table: Honest-but-curious entities

Vehicular Communication Systems



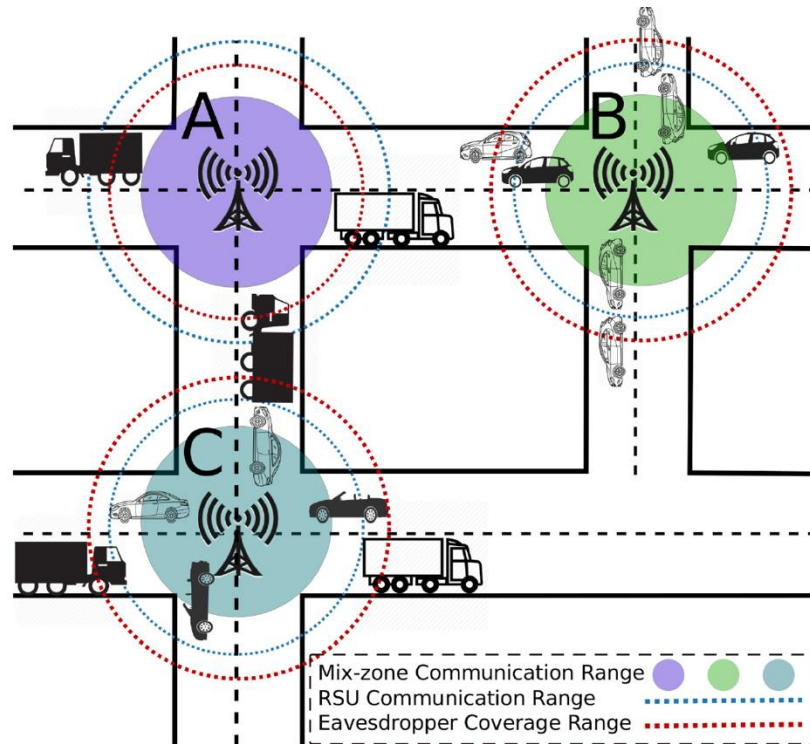
Pseudonymous authentication

- Only identities are unlinkable
- Message content can still reveal extra
- A safety beacon includes, among others, location, direction, velocity, etc.



Mix-zones with decoy traffic

- Chaff pseudonyms used to sign decoy traffic are obtained from the VPKI
- The VPKI proactively constructs a Cuckoo Filter (CF) by including chaff pseudonyms
- Legitimate vehicles efficiently filter out decoy traffic based on the signed CF



[KhodaeiP:J:2021a]

Mix-zones with decoy traffic (cont'd)

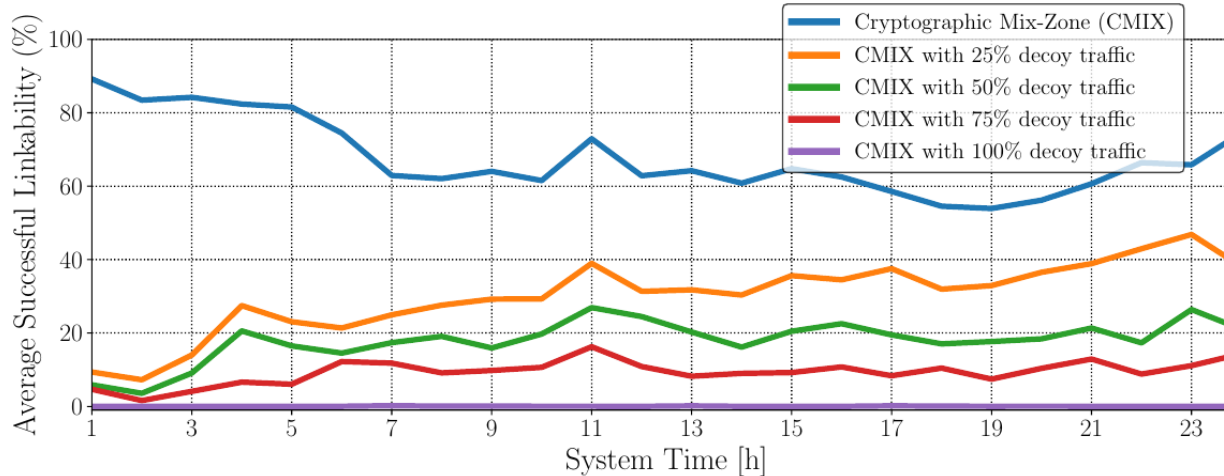
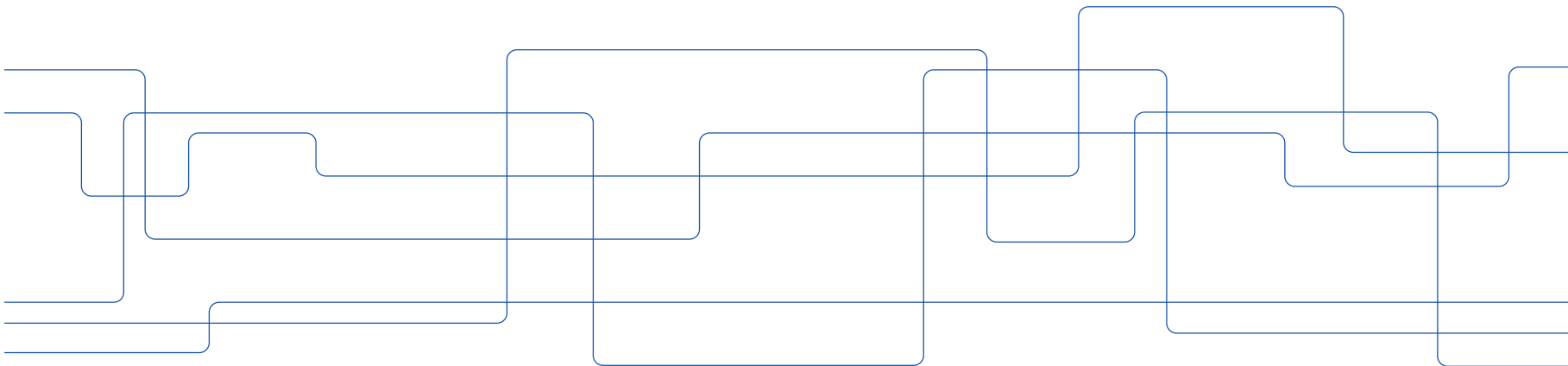


Figure: Syntactic and semantic linking attacks.

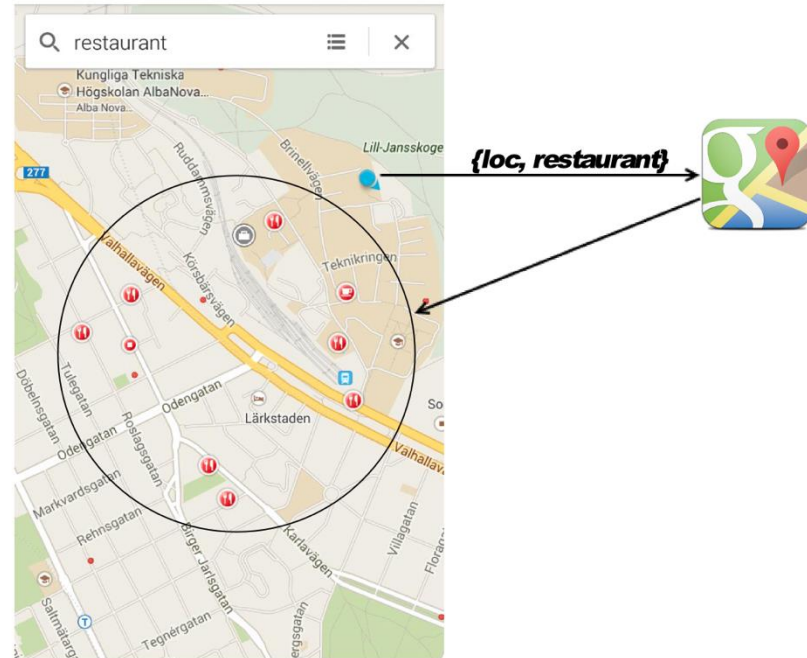
- The probability of linking decreases when the traffic density increases.
- The baseline scheme: successfully linking the pseudonyms with 78% success rates.
- With our scheme, the success rate decreases from $\approx 80\%$ to 40% by introducing decoy traffic (e.g., 25% of vehicles in the mix-zones).

Location-based Services (LBSs)



Privacy exposure in LBS

- LBS queries expose sensitive user information to the honest-but-curious LBS server, including user identity, location and interest

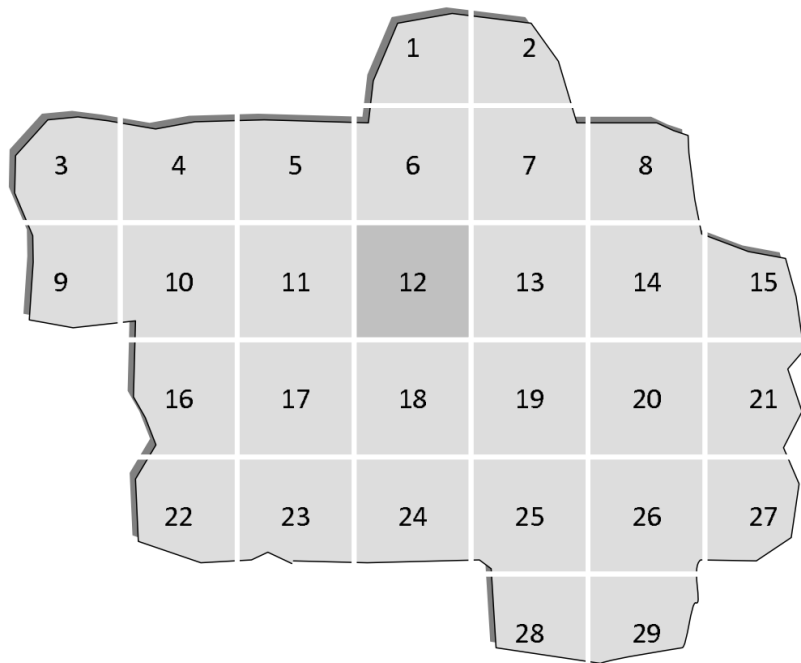




- Ability to accurately locate the user related to the size of the region R, rather than to k

Shokri, Reza, et al. "Unraveling an old cloak: k-anonymity for location privacy." *Proceedings of the 9th annual ACM workshop on Privacy in the electronic society*. 2010.

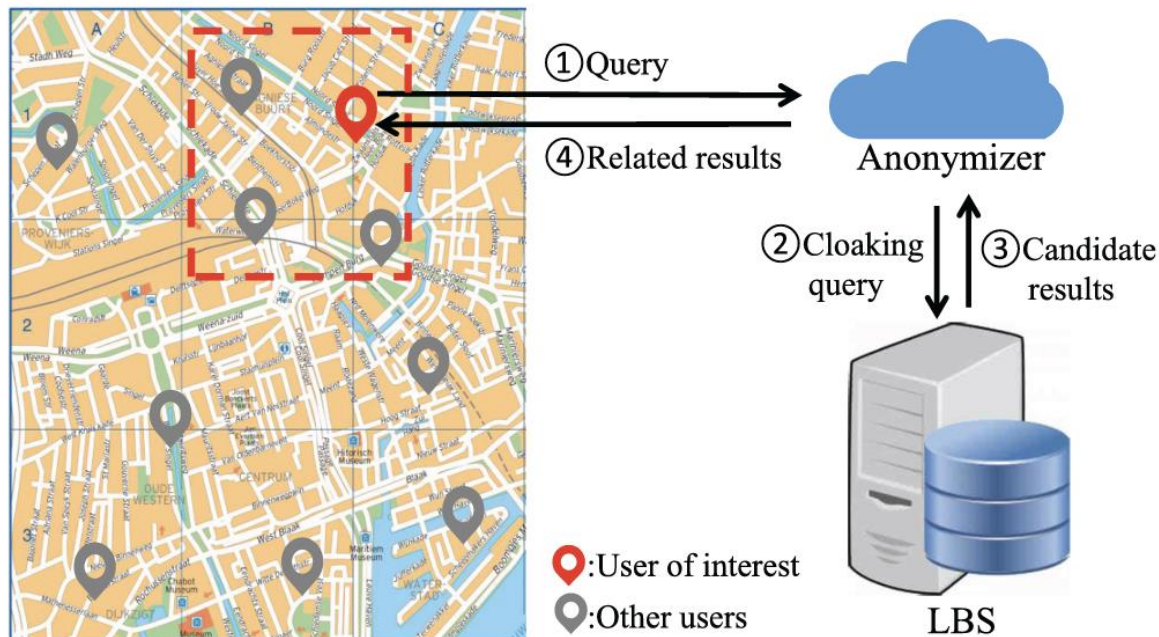
Obfuscation



- The user's actual location is region 12.
- Obfuscation methods will replace 12 with a different location pseudonym r'
- $r' = \{14\}$ in the perturbation method
- $r' = \{12, 15, 26\}$ in the adding dummy regions method
- $r' = \{9, 10, 11, 12, 13, 14, 15\}$ in the reducing precision method

Shokri, Reza, et al. "Quantifying location privacy." IEEE symposium on security and privacy, 2011.

Centralized Anonymizers



Hongbo Jiang, et al. "Location privacy-preserving mechanisms in location-based services: A comprehensive survey." *ACM Computing Surveys (CSUR)* 54.1 (2021): 1-36.



Honest-but-curious Anonymizers

- A single point-of-failure
- Merely shifting the trust to the anonymizers
- Anonymizers themselves could be honest-but-curious
- Not an ultimate solution

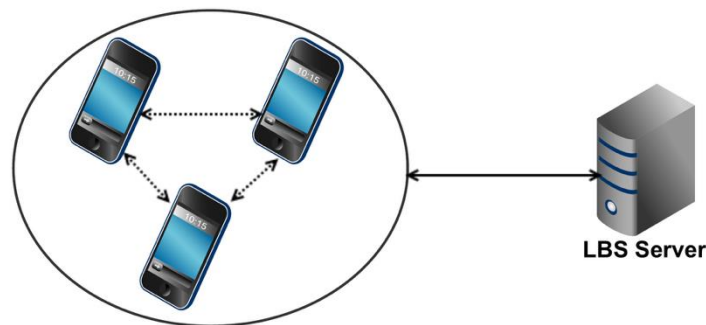
Caching and sharing

- Reduces exposure to LBS
- Cached Point-of-Interest (POI) data can serve future queries and reduce direct queries to the LBS server
- Road-side infrastructure can proactively cache POI data
 - Honest-but-curious
- Nodes can cache information received from the LBS server and pass to neighbors when requested

- B. Liu, W. Zhou, T. Zhu, L. Gao, T. H. Luan, and H. Zhou. 2016. Silence is golden: Enhancing privacy of location-based services by content broadcasting and active caching in wireless vehicular networks. *IEEE transactions on vehicular technology* 65, 12 (2016), 9942–9953
- B. Niu, Q. Li, X. Zhu, G. Cao, and H. Li. 2015. Enhancing privacy through caching in location-based services. In *2015 IEEE INFOCOM*. Kowloon, Hong Kong, 1017–1025.
- R. Shokri, G. Theodorakopoulos, P. Papadimitratos, E. Kazemi, and J.-P. Hubaux. 2014. Hiding in the mobile crowd: Locationprivacy through collaboration. *IEEE Transactions on Dependable and Secure Computing* 11, 3 (2014), 266–279.

Caching and sharing (cont'd)

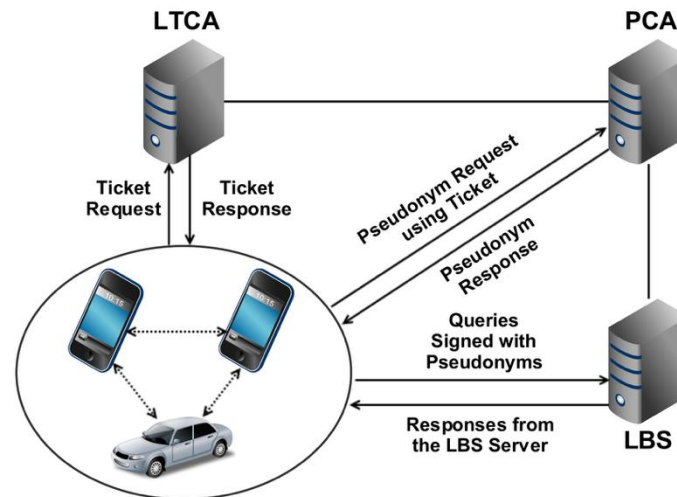
- Objectives
 - Rely on neighbors/peers, without the need of a central anonymizer
 - Cache and share (with neighbors) responses from the LBS
 - Contact the LBS server directly when absolutely necessary
- Problems
 - Significant privacy exposure to curious neighbors
 - Any peer could misbehave
 - Challenge: protect users against adversaries while remaining efficient



R. Shokri, G. Theodorakopoulos, P. Papadimitratos, E. Kazemi, and J.-P. Hubaux. 2014. Hiding in the mobile crowd: Location privacy through collaboration. IEEE Transactions on Dependable and Secure Computing 11, 3 (2014), 266–279.

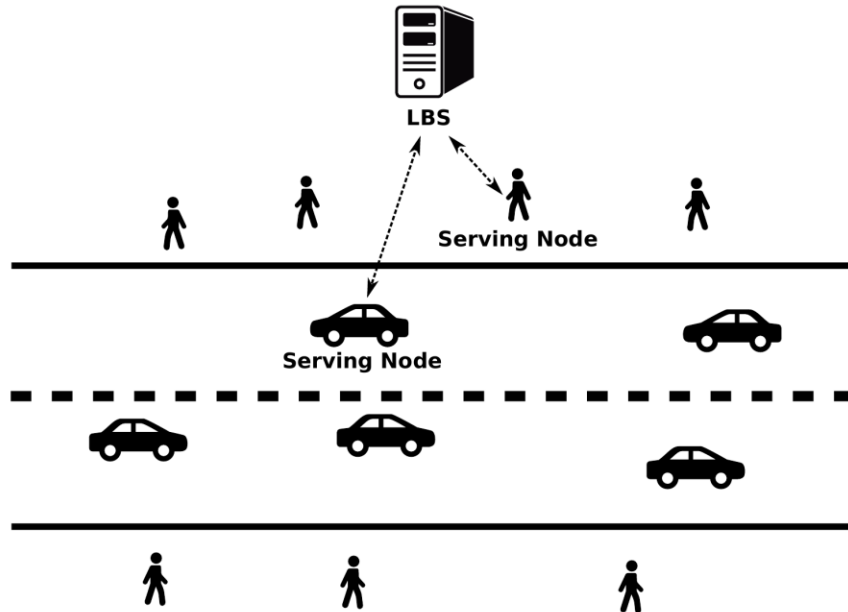
Resilient LBS Privacy

- Ticket and pseudonym acquisition from LTCA and PCA
- Pseudonymous authentication for node-server and peer-to-peer (P2P) communication
- Pseudonym resolution in case of misbehavior with the help of Resolution Authority (RA)



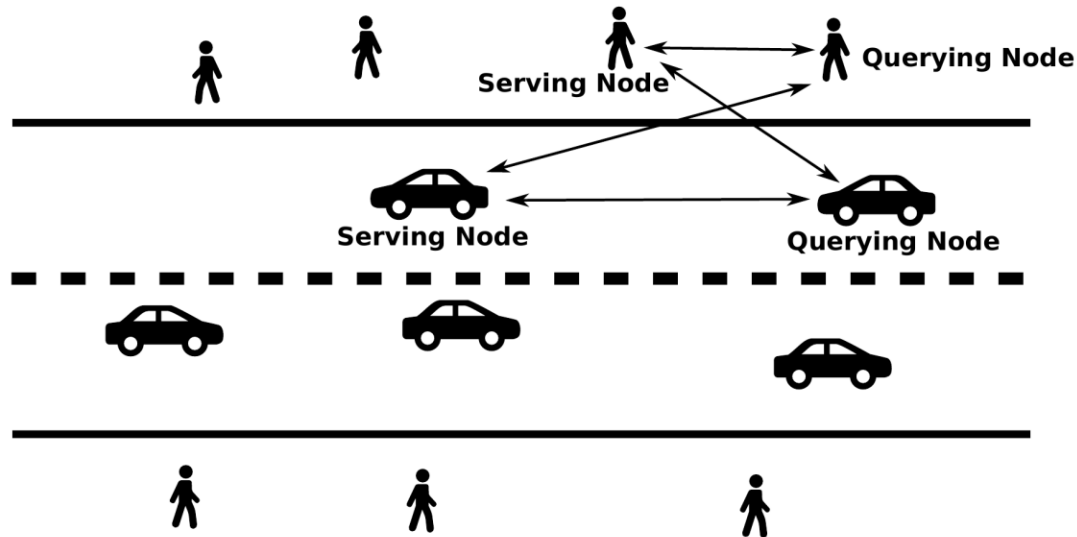
H. Jin and P. Papadimitratos, "Resilient Privacy Protection for Location-Based Services Through Decentralization," ACM Transactions on Privacy and Security (ACM TOPS), vol. 22, no. 4, pp. 21:1-36, September 2019

Resilient LBS Privacy (cont'd)



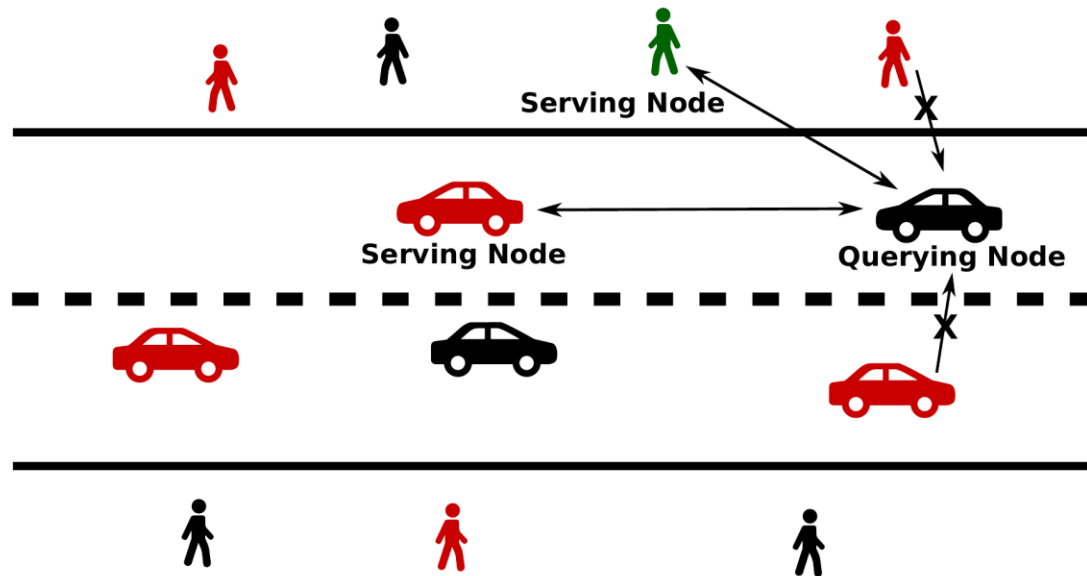
- The whole area divide into equally-sized regions
- Randomly assigned serving nodes proactively cache POI data for the regions they are located in, and serve other nodes
- Announce presence with broadcasted pseudonymously authenticated beacons

Resilient LBS Privacy (cont'd)



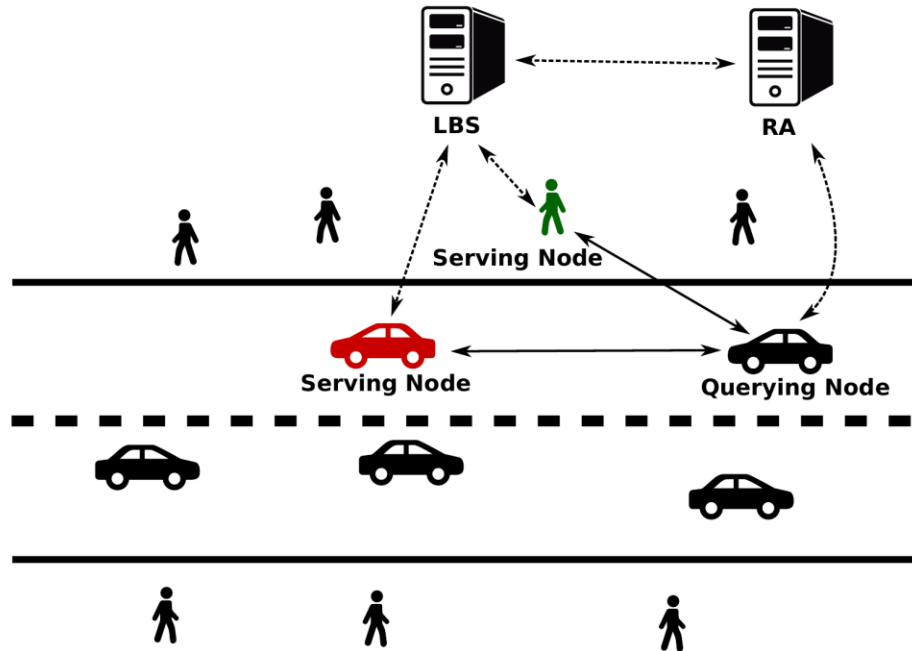
- An interested node listens to beacons and request POI from the serving node

Resilient LBS Privacy (cont'd)



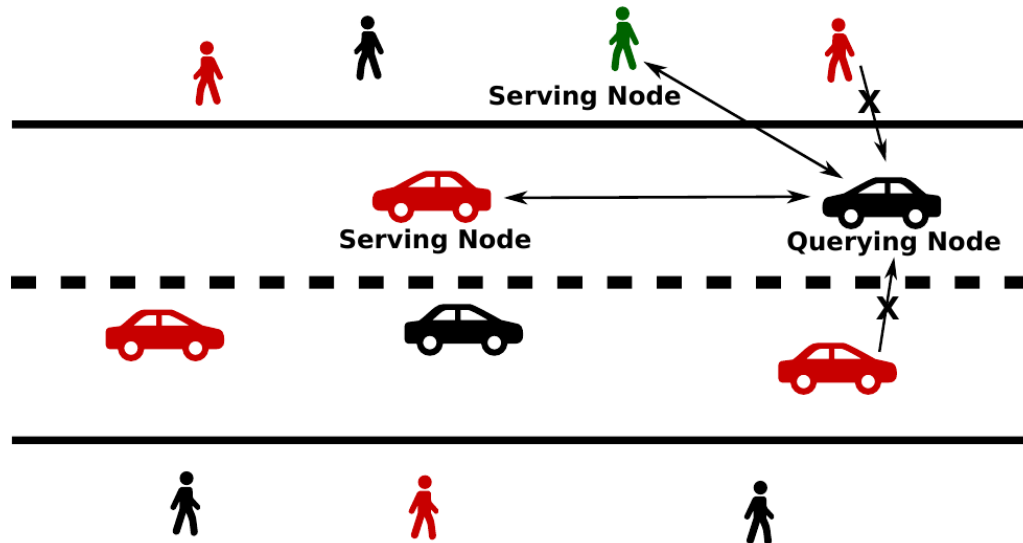
- Only serving nodes will be queried and could respond
- Prevent malicious nodes from actively providing false information

Resilient LBS Privacy (cont'd)



- Can request responses from $N > 1$ serving nodes for cross-checking
- Cross-checking helps identifying the malicious nodes

Resilient LBS Privacy (cont'd)



- Query-response is encrypted with a session key
- Encrypted queries reduce exposure to the neighboring curious nodes

Resilient LBS Privacy (cont'd)

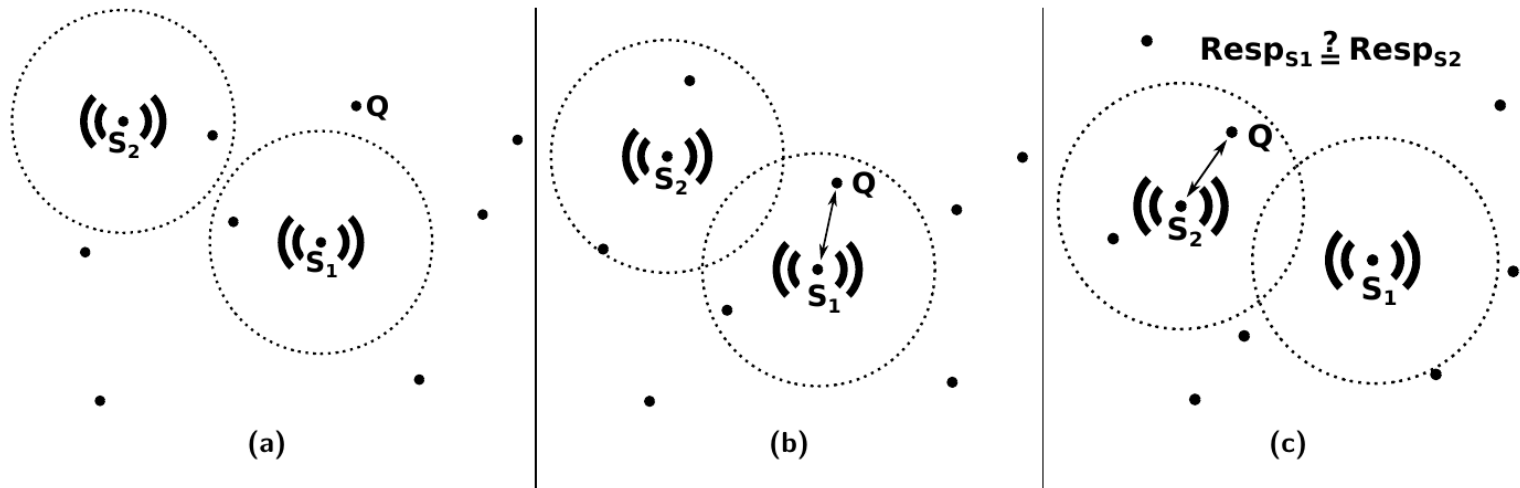


Figure 1: Illustration of peer query and response collection. (a) At t_1 , Q listens for beacons from serving nodes. (b) At t_2 , Q receives beacons broadcasted by S_1 and queries S_1 . (c) At t_3 , Q receives beacons broadcasted by S_2 and queries S_2 .

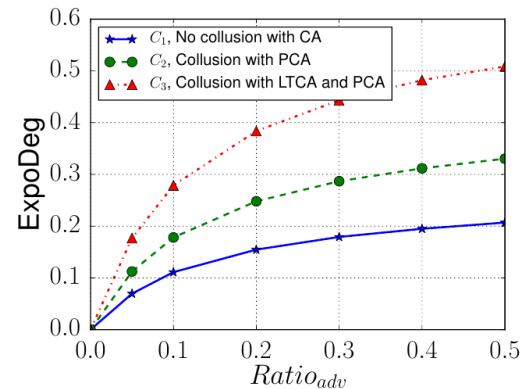
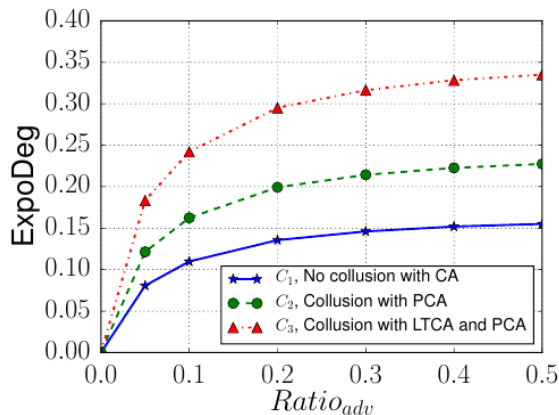
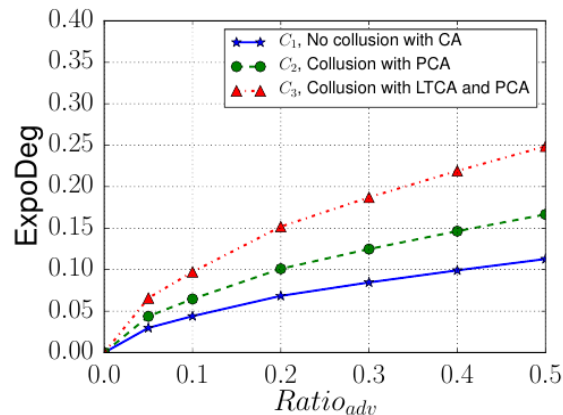
- Probabilistically checks the response with the LBS server if only one response is received
- *Authenticator*, included in each response, being the signature on the combination of query and response
- In case of false response, *Authenticator* proves the misbehavior

Exposure Degree

$$ExpoDeg(Id_{LTC}, C) = \sum_{Id_i \in ID(Id_{LTC}, C)} \frac{T(Id_i)}{T(Id_{LTC})} * \frac{R_H(Id_i)}{R(Id_{LTC})} \quad (1)$$

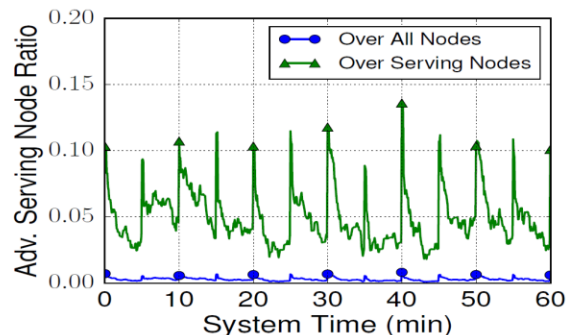
- $ID(Id_{LTC}, C)$: set of identities corresponding to Id_{LTC} exposed to honest-but-curious (possibly colluding) entities
- $T(Id)$: trip duration of a node under identity Id
- $R(Id)$: number of regions the node visits as Id
- $R_H(Id)$: number of visited regions exposed
- $ExpoDeg$: accuracy of reconstructed node trajectories based on recorded node queries, taking into account pseudonymous authentication

Exposure to honest-but-curious entities

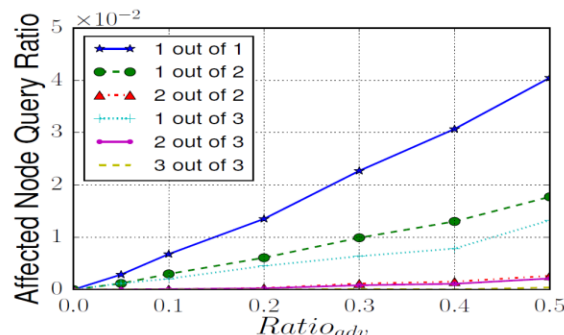


- Exposure degree to colluding passive adversaries as a function of adversary ratio with the resilient LBS privacy scheme (left) with and (middle) without confidentiality for P2P communication, and with MobiCrowd with a cooperation ratio of 50% (right).

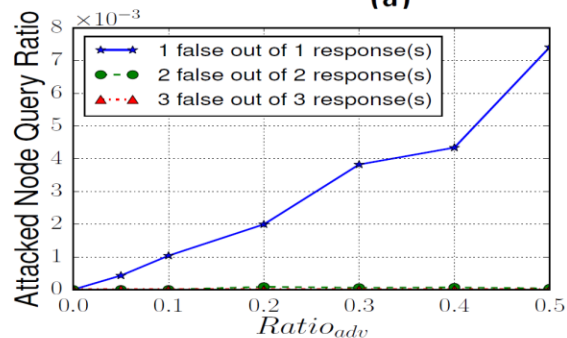
False Response Detection



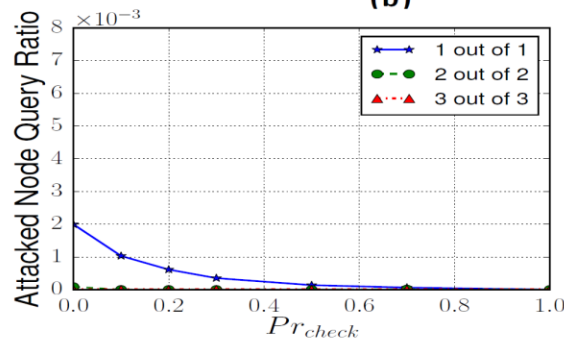
(a)



(b)



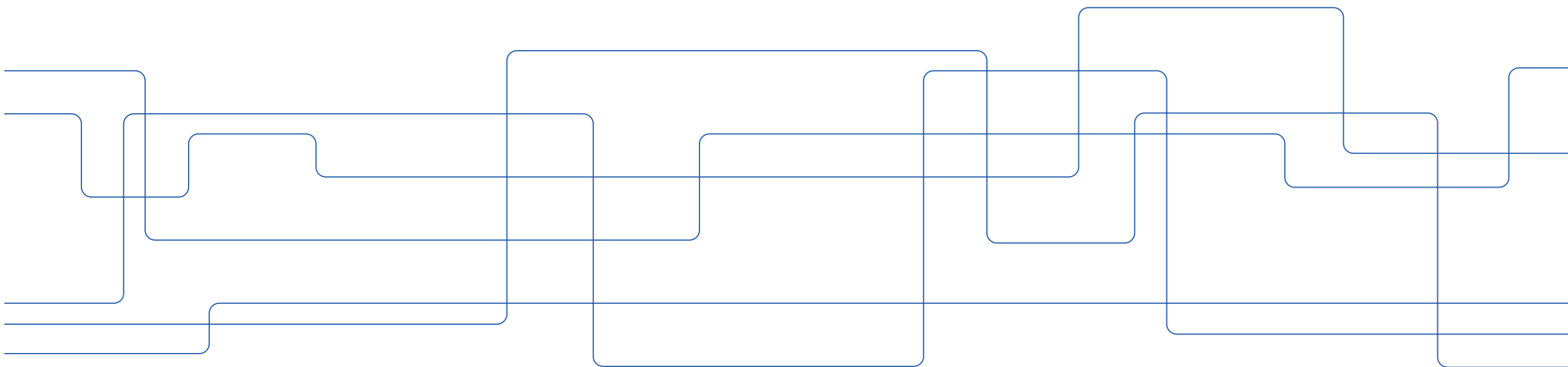
(c)



(d)

Figure: (a) Ratio of malicious serving nodes and (b) affected node query ratio as a function of $\bar{Ratio}_{adv}$. Non-detected attacked node query ratio as a function of (c) $\bar{Ratio}_{adv}$ and (d) Pr_{check} .

Applied Privacy on Mobile Devices



MAC Randomization

draft-ietf-madinas-mac-address-randomization-04

- Globally-unique MAC address make tracking easier based on over the air data transmission
- MAC address exposure:
 - While actively scanning for available networks, the MAC address is used in the Probe Request frames sent by the device (aka IEEE 802.11 STA).
 - Once associated to a given Access Point (AP), the MAC address is used in frame transmission and reception, as one of the addresses used in the address fields of an IEEE 802.11 frame.

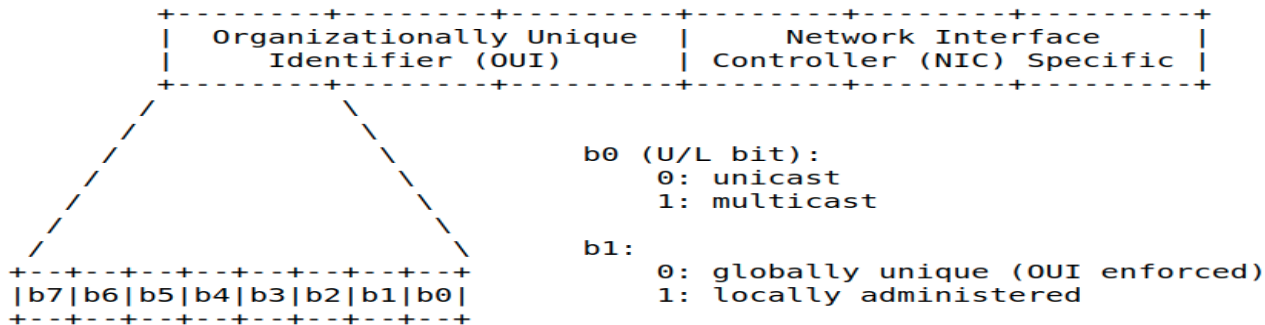
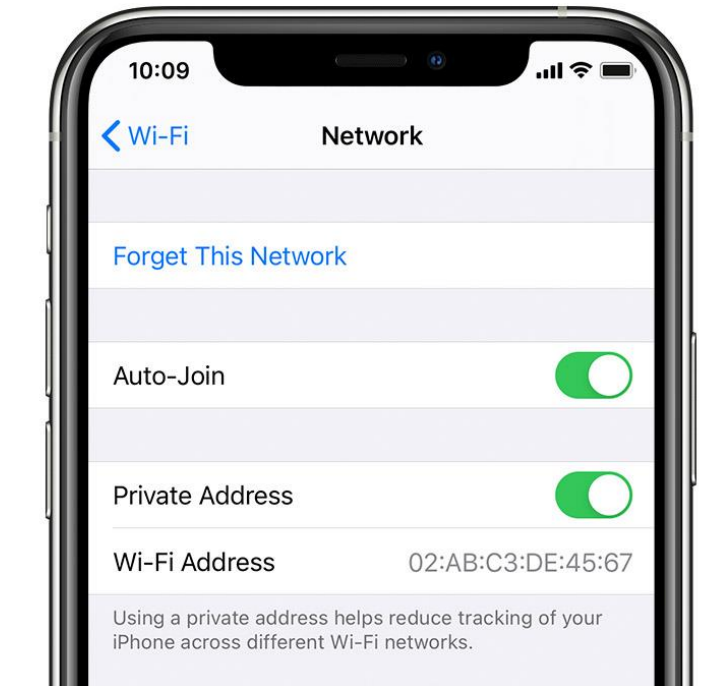


Figure 1: IEEE 802 MAC Address Format

MAC Randomization (cont'd)

Off-the-shelf devices

- Apple uses a different MAC address for each Wi-Fi network
 - Refreshes if disconnected for more than 6 weeks
 - Manually "forget the network" after more than 2 weeks of connection triggers MAC address refresh
- Android generates a persistent randomized MAC address, which remains the same until factory reset
 - Optional non-persistent randomization could refresh MAC address at the start of every connection



MAC Randomization (cont'd)

Why is randomization not sufficient?

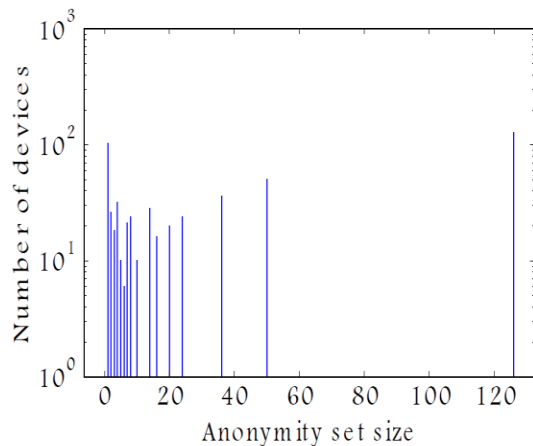
Element	Entropy (bits)			Stability			Affected devices		
	Lab	Station	Sapienza	Lab	Station	Sapienza	Lab	Station	Sapienza
HT capabilities info	3.94	4.74	3.35	96.0%	95.9%	99.6%	90.9%	90.0%	81.1%
Ordered list of tags numbers	4.23	5.24	4.10	93.6%	94.2%	91.2%	100%	100%	100%
Extended capabilities	2.59	2.57	0.064	98.5%	99.4%	99.9%	55.4%	51.3%	0.6%
HT A-MPDU parameters	2.59	2.67	2.54	97.8%	99.1%	99.7%	90.9%	90.0%	81.1%
HT MCS set bitmask	1.49	1.43	1.16	97.6%	99.0%	99.9%	90.9%	90.0%	81.1%
Supported rates	1.18	2.10	1.36	98.2%	95.9%	99.8%	100%	99.9%	100%
Interworking - access net. type	1.08	1.11	0.006	99.6%	99.6%	100.0%	47.5%	46.1%	0.04%
Extended supported rates	1.00	1.77	0.886	98.0%	96.3%	99.4%	99.1%	72.6%	99.7%
WPS UUID	0.878	0.788	0.658	98.2%	99.2%	99.6%	8.4%	5.5%	3.6%
HT extended capabilities	0.654	0.623	0.779	97.8%	98.9%	99.9%	90.9%	90.0%	81.1%
HT TxBeam Forming Cap.	0.598	0.587	0.712	97.8%	98.9%	99.9%	90.9%	90.0%	81.1%
HT Antenna Selection Cap.	0.579	0.576	0.711	98.0%	98.9%	99.9%	90.9%	90.0%	81.1%
Overall	5.48	7.03	5.65	92.5%	90.7%	88.8%	-	-	-

- Analysis of the Information Elements of probe requests in the considered datasets. For each item: the entropy brought by the element, the percentage of devices for which this item is stable over time, and the percentage of devices that include this item in their probe requests.

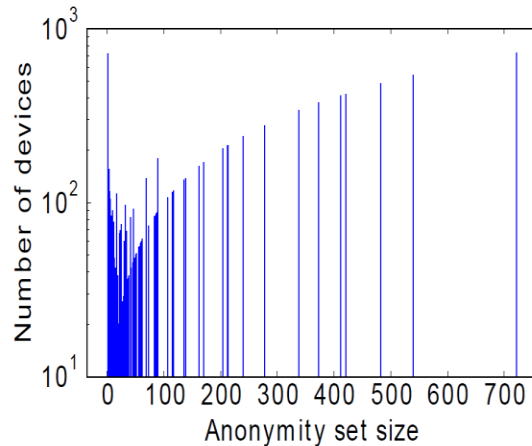
Vanhoef, Mathy, et al. "Why MAC address randomization is not enough: An analysis of Wi-Fi network discovery mechanisms." *Proceedings of the 11th ACM on Asia conference on computer and communications security*. 2016.

MAC Randomization (cont'd)

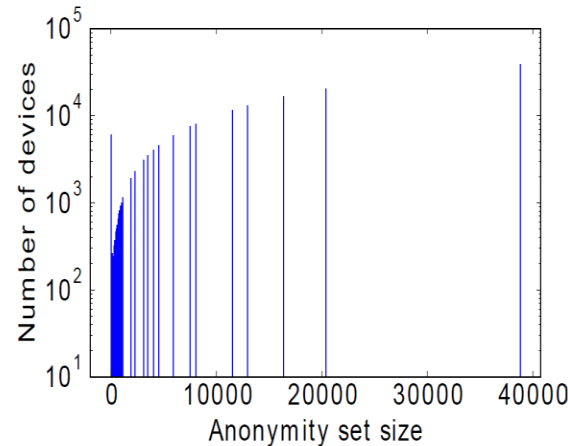
Why is randomization not sufficient?



(a) Lab



(b) Train-station



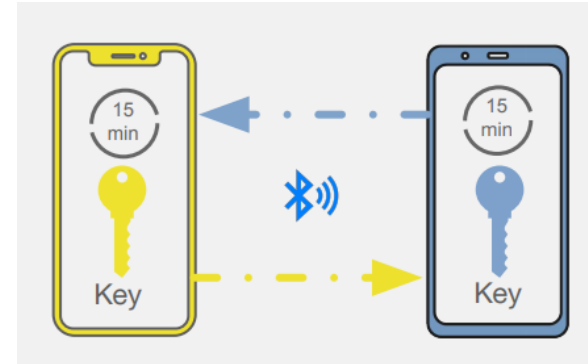
(c) Sapienza

Figure: Number of devices sharing the same IE fingerprint with a group (i.e., anonymity set) of varying size.

Contact Tracing for Covid-19

Exposure Notification by Apple and Google

- Phones broadcast beacons with random Bluetooth identifiers
- Random Bluetooth identifiers rotate every 10-20 minutes, to help prevent tracking
- If a user chooses to report a positive diagnosis of COVID-19
 - Most recent keys to their Bluetooth beacons added to the positive diagnosis list
 - List shared by the public health authority
 - Other users in contact with those beacons can be alerted

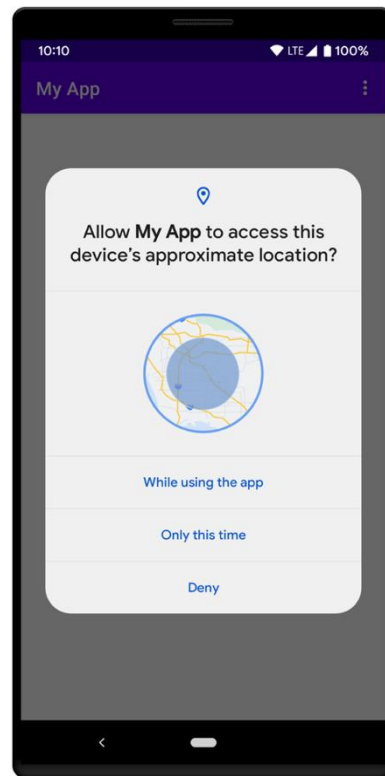


<https://covid19.apple.com/contacttracing>

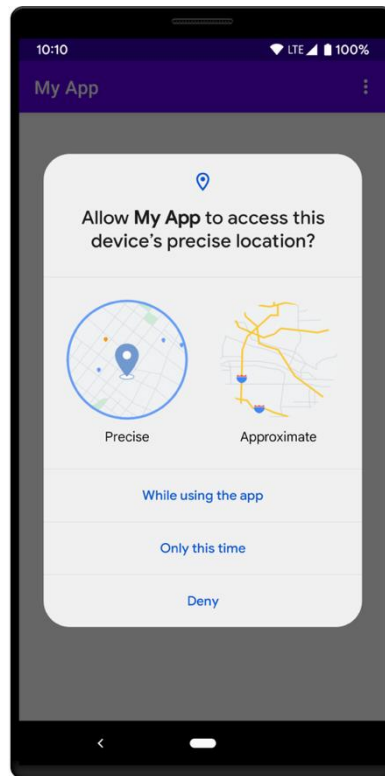
Android Location Privacy

- On Android 12 (API level 31) or higher, users can request that your app retrieve only approximate location information
- **Precise:** Allows your app to get precise location information.
- **Approximate:** Allows your app to get only approximate location information.
 - Essentially obfuscation
 - Only a first step towards location privacy; advanced inference attacks still possible

<https://developer.android.com/training/location/permissions>



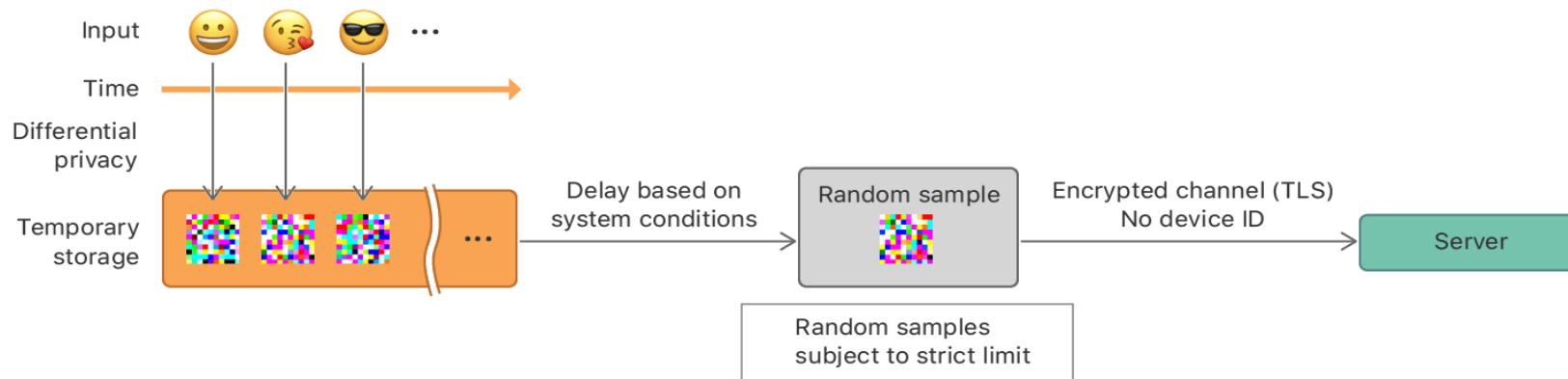
[ACCESS_COARSE_LOCATION]



[ACCESS_FINE_LOCATION]

Data Collection by Apple

- Data is immediately privatized via local **differential privacy**
- After a delay based on device conditions, the system randomly samples from the differentially private records based on a daily transmission limit
- Records do not include device identifiers or timestamps of when events were generated
- The communication between a device and the server is encrypted using TLS



<https://machinelearning.apple.com/research/learning-with-privacy-at-scale>

Differential Privacy

- Mathematical definition of privacy
- The output of a differentially private analysis will be roughly the same, whether or not you contribute your data
- By looking at the output, one cannot tell whether any individual's data was included in the original dataset or not



Analysis $\mathcal{M}$ satisfies differential privacy if...

For all D_1 and D_2 which **differ in one individual's data**...

Answer **A** and answer **B** are **indistinguishable**

Probability of seeing output O on input D_1

Probability of seeing output O on input D_2

$$\frac{\Pr[\mathcal{M}(D_1) \in O]}{\Pr[\mathcal{M}(D_2) \in O]} \leq e^\epsilon$$

Indistinguishability:
bounded ratio of probabilities

<https://www.nist.gov/blogs/cybersecurity-insights/differential-privacy-privacy-preserving-data-analysis-introduction-our>



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